

SUMATHI K N

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PROFESSIONAL SUMMARY

BCA graduate (CGPA 9.83/10) with a passion for technology, innovation and real-world problem solving. Skilled in Python, Flask, HTML, CSS, SQL, Bootstrap and pandas. Completed a Full Stack Development training internship at Edu Phoenix and independently built a Blood Donation Portal web application using Flask and MySQL. Independently built and shipped two web applications — a REST API driven university portal and a blood donation management system with measurable results. Coursework in Machine Learning and Cybersecurity. Proficient in Linux, Git, and Postman. Seeking a junior developer role to deliver real-world impact.

TECHNICAL SKILLS

Frontend: HTML | CSS

Backend: Python | PHP

Frameworks: Flask | Bootstrap

ML: scikit-learn | pandas | numpy | matplotlib | seaborn | Google Colab

ML Algorithms: Decision Tree | KNN | Random Forest | ANOVA F-test Feature Selection

ML Techniques: GridSearchCV | StratifiedKFold | 5-fold Cross-Validation | ROC-AUC | Confusion Matrix

Cybersecurity: Intrusion Detection Systems (IDS) | NSL-KDD Dataset | Attack Classification

Database: MySQL | SQL | Postresql |

Tools & IDEs: Visual Studio Code | PyCharm | Turbo | MySQL Workbench

INTERNSHIP

Full Stack Development Internship (Training-Based) | Edu Phoenix — Academic
/ Training Program

02/2025 – 04/2025

- Completed hands-on training in full stack development using Python, Flask, and MySQL.
- Worked on practice modules simulating real-world web development tasks covering frontend-backend integration.
- Understood and applied frontend-backend integration concepts through guided examples and mentor-led sessions.
- Gained practical exposure to form handling, data validation, routing in Flask, and MySQL database operations.

PROJECTS

IDS Extended Research Study — ML for Cybersecurity | Academic Project

2026

Tech Stack: Python | scikit-learn | pandas | numpy | seaborn | matplotlib | Google Colab | NSL-KDD

- Extended Pathak & Pathak (2020) IDS paper by adding Random Forest as a third classifier alongside Decision Tree and KNN.
- Applied GridSearchCV with StratifiedKFold (5-fold) to tune hyperparameters — DT max_depth=10, KNN k=5, Random Forest 100 estimators.
- Achieved 99.43% accuracy and ROC-AUC 0.987 with Random Forest on the NSL-KDD benchmark (125,973 training records, 4 attack classes: DoS, Probe, R2L, U2R).
- Added ROC-AUC score and 5-fold cross-validation — CV standard deviation below 0.32% across all models confirming statistical stability.
- Built and delivered: IEEE-format extended paper (Word), 12-slide PowerPoint, Google Colab notebook with prompt/code/inference, and an interactive HTML report.
- Implemented normalised confusion matrix heatmaps and per-class ROC curves (One-vs-Rest) using seaborn and matplotlib.

University Inquiry Management System | Academic Project

Tech Stack: | Django | PostgreSQL | REST API | Bootstrap | Postman |

2026

- Designed and built a full-stack university portal with 5 PostgreSQL tables (Course, Faculty, Hostel, Transport, Inquiry) using foreign key constraints and cascade delete rules.
- Exposed 5 REST API endpoints returning structured JSON; validated all routes via Postman — 200 OK across every endpoint at submission with zero failures.

- Built role-gated admin dashboard (Django auth, staff/superuser only) aggregating inquiries by category — blocked all unauthorised access during testing.
- Enforced server-side validation on 6+ form fields including regex phone check — reduced malformed submissions to 0 across all test runs.

Blood Donation Portal | Personal Project

2025

Tech Stack: Python | Flask | MySQL | HTML | CSS | Bootstrap

- Developed a full-stack web application using Python Flask and MySQL for managing blood donors and requests.
- Implemented donor registration, blood availability tracking, and an admin dashboard with full CRUD operations.
- Admin can view, manage, and generate reports of registered donors and blood requests with structured database design.
- Focused on backend logic, server-side form validations, and structured database schema design for reliability and usability.
- Received Project Certificate upon successful completion and demonstration of the working application.

EDUCATION

Master of Computer Applications (MCA) | Chanakya University, Bengaluru | 2025 – 2027

June 2025

CGPA: 9.47 / 10 (1st Sem) | Specialisation: Cybersecurity | Courses: Machine Learning, Full-Stack Development, DBMS

Bachelor of Computer Applications (BCA) | Best Degree College, Chikkaballapur
| Bengaluru North University

CGPA: 9.83 / 10 | Location: Chikkaballapur, Karnataka

- Core courses: Python, Web Development, PHP, MySQL DBMS, C Programming

CERTIFICATIONS & TRAINING

IDS Extended Research — ML & Cybersecurity | Self-Directed Academic Project

2026

- Conducted a full extended ML research study on Intrusion Detection using NSL-KDD; produced IEEE paper, PowerPoint, Colab notebook, and HTML report.
- Demonstrated hands-on proficiency in scikit-learn, GridSearchCV, ROC-AUC, 5-fold CV, seaborn heatmaps and multi-class classification.

Full Stack Development Training — Project Certificate | Edu Phoenix Solutions

Feb – Apr 2025

- Completed hands-on training in full stack web development covering Python, Flask, MySQL, and frontend technologies.
- Awarded Project Certificate on successful completion of Blood Donation Portal web application.

SOFT SKILLS

Adaptability | Friendly Nature | Never Repeat Mistakes | Report Generation | Quick Learner | Team Collaboration

LANGUAGES

Kannada | English